

Lab 7: FFmpeg (Audio & Video Processing)

Objective: To learn how to process multimedia files using FFmpeg, including conversion, compression, editing, filtering, and optimization of audio and video.

Introduction: FFmpeg is a powerful open-source multimedia framework used for handling video, audio, and other multimedia files. It supports a wide range of operations such as format conversion, compression, trimming, filtering, and streaming.

Tools/Software Used:

- FFmpeg
- Command Line Interface (CMD / Terminal)

1. Basic Overview of FFmpeg

FFmpeg is a powerful open-source tool used for:

- Converting audio/video formats
- Editing, compressing, streaming
- Applying filters and effects

Basic syntax: `ffmpeg -i input.mp4 output.avi`

Video Tasks

1. Conversion

MP4 → AVI

```
ffmpeg -i input.mp4 output.avi
```

MP4 → MKV

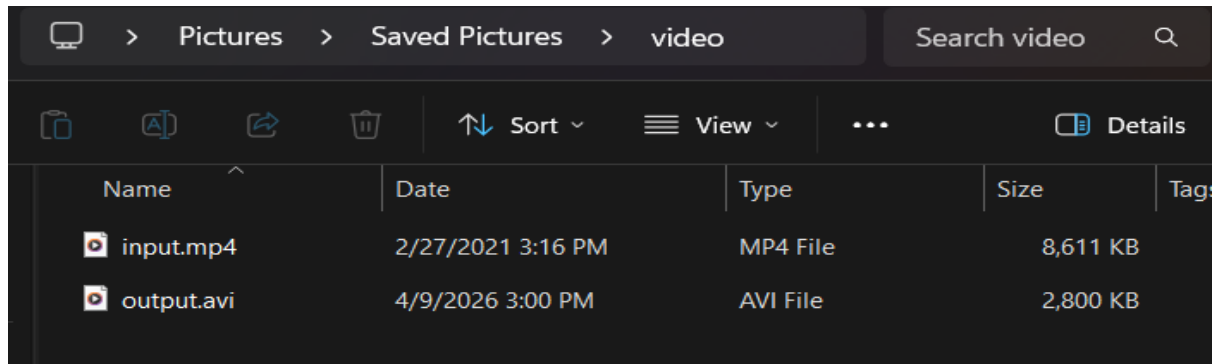
```
ffmpeg -i input.mp4 output.mkv
```

Output

```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp4 output.avi
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFmpeg developers
  built with gcc 15.2.0 (crosstool-NG 1.28.0.23_185f348)
  configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threads --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enable-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disable-libxcb --disable-xlib --disable-libpulse --enable-opencore-amrnb --enable-gmp --enable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libaribb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libdav1d --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ffnvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkvazaar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjxl --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist --enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enable-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enable-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband --enable-libschannel --enable-sdl2 --enable-libsndio --enable-libsoxr --enable-libsrt --enable-libsvtav1 --enable-libtesseract --enable-libuavs3d --disable-libdrm --enable-vaapi --enable-libvidstab --enable-libvvc --enable-whisper --enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enable-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxflags= --extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags= --cc=x86_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gcc-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm --extra-version=20260407
  libavutil      60. 29.100 / 60. 29.100
  libavcodec     62. 29.101 / 62. 29.101
  libavformat    62. 13.101 / 62. 13.101
  libavdevice    62.  4.100 / 62.  4.100
  libavfilter    11. 15.101 / 11. 15.101
  libswscale     9.  7.100 /  9.  7.100
  libswresample  6.  4.100 /  6.  4.100
Input #0, mov,mp4,m4a,3gp,3g2,mj2, from 'input.mp4':
  frame= 17 fps=0.0 q=31.0 size= 256KiB time=00:00:01.24 bitrate=1679.1k
  frame= 77 fps= 74 q=31.0 size= 768KiB time=00:00:03.81 bitrate=1648.3k
  frame= 131 fps= 85 q=31.0 size= 1280KiB time=00:00:06.07 bitrate=1726.6k
  frame= 190 fps= 92 q=31.0 size= 1792KiB time=00:00:08.54 bitrate=1718.0k
  frame= 242 fps= 94 q=24.8 size= 2048KiB time=00:00:10.84 bitrate=1546.4k
[out#0/avi @ 0000025bb9dcb4c0] video:2571KiB audio:200KiB subtitle:0KiB other streams:0KiB global headers:0KiB muxing overhead: 1.064153%
  frame= 305 fps=101 q=31.0 Lsize= 2800KiB time=00:00:12.73 bitrate=1801.0 kbits/s speed= 4.2x elapsed=0:00:03.03

C:\Users\Dell\Pictures\Saved Pictures\video>
```



2. Compression (reduce file size)

```
ffmpeg -i input.mp4 -vcodec libx264 -crf 28 output.mp4
```

- Lower CRF = better quality (18–28 recommended)

Output

```
C:\Users\ DELL\ Pictures\ Saved Pictures\ video>ffmpeg -i input.mp4 -vcodec libx264 -crf 28 output.mp4
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFmpeg developers
  built with gcc 15.2.0 (crosstool-NG 1.28.0.23_185f348)
  configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threads --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enable-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disable-libxcb --disable-xlib --disable-libpulse --enable-opencl --enable-gmp --enable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libarib24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libdav1d --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ffnvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkvazaar --enable-libribcaption --enable-libass --enable-libbluray --enable-libjxl --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist --enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enable-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enable-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband --enable-schannel --enable-sdl2 --enable-libsnappy --enable-libsoxr --enable-libsrt --enable-libsvtav1 --enable-libtwolame --enable-libuavs3d --disable-libbdrn --enable-vaapi --enable-libvidstab --enable-libvvc --enable-whisper --enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enable-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxflags=-extra-libs=lgomp --extra-ldflags=-pthread --extra-ldexeflags=-cc=x86_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gcc-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm --extra-version=20260407
  libavutil      60. 29.100 / 60. 29.100
  libavcodec     62. 29.101 / 62. 29.101
  libavformat    62. 13.101 / 62. 13.101
  libavdevice    62.  4.100 / 62.  4.100
  libavfilter    11. 15.101 / 11. 15.101
  libswscale     9.  7.100 / 9.  7.100
  libswresample  6.  4.100 / 6.  4.100
Input #0, mov,mp4,m4a,3gp,3g2,mj2, from 'input.mp4':
```

```

frame= 101 fps= 21 q=33.0 size=    768KiB time=00:00:07.46 bitrate= 842.7k
frame= 122 fps= 23 q=33.0 size=   1024KiB time=00:00:08.68 bitrate= 966.2k
frame= 141 fps= 24 q=33.0 size=   1024KiB time=00:00:09.47 bitrate= 885.7k
frame= 161 fps= 26 q=33.0 size=   1280KiB time=00:00:10.26 bitrate=1021.9k
frame= 177 fps= 26 q=33.0 size=   1280KiB time=00:00:11.05 bitrate= 948.9k
frame= 189 fps= 25 q=33.0 size=   1280KiB time=00:00:11.54 bitrate= 908.6k
frame= 205 fps= 26 q=33.0 size=   1536KiB time=00:00:12.20 bitrate=1031.2k
frame= 221 fps= 26 q=33.0 size=   1536KiB time=00:00:12.58 bitrate= 999.7k
frame= 227 fps= 25 q=33.0 size=   1536KiB time=00:00:09.38 bitrate=1340.9k
frame= 263 fps= 27 q=33.0 size=   2048KiB time=N/A bitrate=N/A speed=N/A e
frame= 287 fps= 28 q=-1.0 size=   2048KiB time=N/A bitrate=N/A speed=N/A e
[out#0/mp4 @ 000001760b0dc5c0] video:2417KiB audio:4KiB subtitle:0KiB other
streams:0KiB global headers:0KiB muxing overhead: 0.414087%
frame= 305 fps= 30 q=-1.0 Lsize=   2431KiB time=00:00:12.73 bitrate=1563.6
kb/s speed=1.24x elapsed=0:00:10.30
[libx264 @ 000001760d67f040] frame I:3      Avg QP:22.18   size: 52934
[libx264 @ 000001760d67f040] frame P:77      Avg QP:25.81   size: 19144
[libx264 @ 000001760d67f040] frame B:225      Avg QP:28.38   size: 3741
[libx264 @ 000001760d67f040] consecutive B-frames:  1.0%  1.3%  2.0% 95.7%
[libx264 @ 000001760d67f040] mb I  I16..4: 49.4% 42.6%  8.1%
[libx264 @ 000001760d67f040] mb P  I16..4:  4.4%  5.0%  0.2%  P16..4: 42.6%
 6.8%  3.6%  0.0%  0.0%   skip:37.3%
[libx264 @ 000001760d67f040] mb B  I16..4:  0.2%  0.2%  0.0%  B16..8: 32.2%
 0.7%  0.1%  direct: 0.2%   skip:66.3%  L0:37.9% L1:61.0% BI: 1.1%
[libx264 @ 000001760d67f040] 8x8 transform intra:49.3% inter:82.5%
[libx264 @ 000001760d67f040] coded y,uvDC,uvAC intra: 17.7% 53.0% 8.9% inter
: 3.1% 8.8% 0.4%
[libx264 @ 000001760d67f040] i16 v,h,dc,p: 43% 23% 16% 18%
[libx264 @ 000001760d67f040] i8 v,h,dc,ddl,ddr,vr,hd,vl,hu: 15% 13% 50%  3%
 4%  5%  4%  3%  3%
[libx264 @ 000001760d67f040] i4 v,h,dc,ddl,ddr,vr,hd,vl,hu: 23% 19% 15%  5%
 9% 11%  7%  5%  5%
[libx264 @ 000001760d67f040] i8c dc,h,v,p: 58% 20% 17%  5%
[libx264 @ 000001760d67f040] Weighted P-Frames: Y:0.0% UV:0.0%
[libx264 @ 000001760d67f040] ref P L0: 66.1% 12.2% 17.3%  4.5%
[libx264 @ 000001760d67f040] ref B L0: 93.0%  5.8%  1.2%
[libx264 @ 000001760d67f040] ref B L1: 97.1%  2.9%
[libx264 @ 000001760d67f040] kb/s:1556.24
[aac @ 000001760d351500] Qavg: 65536.000

```

× + > Pictures > Saved Pictures > video Search video				
Sort View Details				
Name	Date	Type	Size	Ta
 input.mp4	2/27/2021 3:16 PM	MP4 File	8,611 KB	
 output.avi	4/9/2026 3:00 PM	AVI File	2,800 KB	
 output.mp4	4/9/2026 3:04 PM	MP4 File	2,431 KB	

3. Adjust Video Quality

Change bitrate

```
ffmpeg -i input.mp4 -b:v 1000k output.mp4
```

Change resolution

```
ffmpeg -i input.mp4 -vf scale=1280:720 output.mp4
```

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp4 -b:v 1000k o
utput.mp4
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
  built with gcc 15.2.0 (crosstool-NG 1.28.0.23_185f348)
  configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-openc1 --enable-gmp --e
nable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
ibb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkvaza
ar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enab
le-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enabl
e-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband -
--enable-schannel --enable-sdl2 --enable-libsnpappy --enable-libsoxr --enable-
libsrt --enable-libsvtav1 --enable-libtwolame --enable-libuavs3d --disable-l
ibdrm --enable-vaapi --enable-libvidstab --enable-libvenc --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs= --extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags= --cc=x8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm -
--extra-version=20260407
  libavutil      60. 29.100 / 60. 29.100
  libavcodec     62. 29.101 / 62. 29.101
  libavformat    62. 13.101 / 62. 13.101
  libavdevice    62.  4.100 / 62.  4.100
  libavfilter    11. 15.101 / 11. 15.101
  libswscale      9.  7.100 /  9.  7.100
  libswresample  6.  4.100 /  6.  4.100
Input #0, mov,mp4,m4a,3gp,3g2,mj2, from 'input.mp4':
  Metadata:
    major_brand      : mp42
```

```

frame= 66 fps= 21 q=39.0 size= 256KiB time=00:00:06.61 bitrate= 317.2k
frame= 86 fps= 23 q=38.0 size= 256KiB time=00:00:06.99 bitrate= 299.8k
frame= 110 fps= 26 q=37.0 size= 512KiB time=00:00:08.23 bitrate= 509.4k
frame= 133 fps= 28 q=37.0 size= 512KiB time=00:00:09.19 bitrate= 456.2k
frame= 157 fps= 30 q=36.0 size= 512KiB time=00:00:10.60 bitrate= 395.6k
frame= 181 fps= 31 q=36.0 size= 768KiB time=00:00:11.73 bitrate= 536.2k
frame= 201 fps= 32 q=36.0 size= 768KiB time=00:00:12.24 bitrate= 513.8k
frame= 220 fps= 32 q=36.0 size= 768KiB time=00:00:09.09 bitrate= 692.0k
frame= 236 fps= 32 q=37.0 size= 1024KiB time=N/A bitrate=N/A speed=N/A e
frame= 275 fps= 35 q=37.0 size= 1280KiB time=N/A bitrate=N/A speed=N/A e
[out#0/mp4 @ 0000018a37f0e6c0] video:1507KiB audio:4KiB subtitle:0KiB other
streams:0KiB global headers:0KiB muxing overhead: 0.663695%
frame= 305 fps= 37 q=-1.0 Lsize= 1520KiB time=00:00:12.73 bitrate= 977.9
kb/s speed=1.56x elapsed=0:00:08.18
[libx264 @ 0000018a3a405040] frame I:3 Avg QP:25.51 size: 40495
[libx264 @ 0000018a3a405040] frame P:77 Avg QP:30.00 size: 11351
[libx264 @ 0000018a3a405040] frame B:225 Avg QP:33.58 size: 2430
[libx264 @ 0000018a3a405040] consecutive B-frames: 1.0% 1.3% 2.0% 95.7%
[libx264 @ 0000018a3a405040] mb I I16..4: 54.2% 38.8% 7.0%
[libx264 @ 0000018a3a405040] mb P I16..4: 3.3% 4.0% 0.1% P16..4: 32.9%
4.0% 1.9% 0.0% 0.0% skip:53.9%
[libx264 @ 0000018a3a405040] mb B I16..4: 0.1% 0.2% 0.0% B16..8: 26.0%
0.3% 0.0% direct: 0.1% skip:73.4% L0:35.9% L1:63.7% BI: 0.4%
[libx264 @ 0000018a3a405040] final ratefactor: 31.45
[libx264 @ 0000018a3a405040] 8x8 transform intra:49.5% inter:85.3%
[libx264 @ 0000018a3a405040] coded y,uvDC,uvAC intra: 15.2% 43.7% 6.5% inter
: 1.5% 4.1% 0.2%
[libx264 @ 0000018a3a405040] i16 v,h,dc,p: 49% 22% 14% 16%
[libx264 @ 0000018a3a405040] i8 v,h,dc,ddl,ddr,vr,hd,vl,hu: 12% 10% 56% 3%
4% 5% 4% 3% 3%
[libx264 @ 0000018a3a405040] i4 v,h,dc,ddl,ddr,vr,hd,vl,hu: 24% 19% 17% 5%
8% 11% 7% 5% 4%
[libx264 @ 0000018a3a405040] i8c dc,h,v,p: 65% 17% 14% 4%
[libx264 @ 0000018a3a405040] Weighted P-Frames: Y:0.0% UV:0.0%
[libx264 @ 0000018a3a405040] ref P L0: 64.3% 14.5% 17.4% 3.8%
[libx264 @ 0000018a3a405040] ref B L0: 94.0% 4.9% 1.1%
[libx264 @ 0000018a3a405040] ref B L1: 97.8% 2.2%
[libx264 @ 0000018a3a405040] kb/s:969.83
[aac @ 0000018a39aee440] Qavg: 65536.000

```

> Pictures > Saved Pictures > video Search video				
Sort View Details				
Name	Date	Type	Size	Ta
 input.mp4	2/27/2021 3:16 PM	MP4 File	8,611 KB	
 output.mp4	4/9/2026 3:09 PM	MP4 File	1,521 KB	

4. Trim Video

ffmpeg -i input.mp4 -ss 00:00:10 -to 00:00:30 -c copy output.mp4

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp4 -ss 00:00:10
-to 00:00:30 -c copy output.mp4
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
  built with gcc 15.2.0 (crosstool-NG 1.28.0.23_185f348)
  configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-openc1 --enable-gmp --e
nable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
ibb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkvaza
ar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enab
le-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enabl
e-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband -
--enable-schannel --enable-sdl2 --enable-libsndio --enable-libsoxr --enable-
libsrt --enable-libsvtav1 --enable-libtwolame --enable-libuavs3d --disable-l
ibdrm --enable-vaapi --enable-libvidstab --enable-libvenc --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs= --extra-libs=lgomp --extra-ldflags=-pthread --extra-ldexeflags= --cc=x8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm -
--extra-version=20260407
  libavutil      60. 29.100 / 60. 29.100
  libavcodec     62. 29.101 / 62. 29.101
  libavformat    62. 13.101 / 62. 13.101
  libavdevice    62.  4.100 / 62.  4.100
  libavfilter    11. 15.101 / 11. 15.101
  libswscale      9.  7.100 /  9.  7.100
  libswresample  6.  4.100 /  6.  4.100
Input #0, mov,mp4,m4a,3gp,3g2,mj2, from 'input.mp4':
```

```
  handler_name    : E-SHAKN Audio Handler
Press [q] to stop, [?] for help
[out#0/mp4 @ 000002973e17ad80] video:535KiB audio:84KiB subtitle:0KiB other
streams:0KiB global headers:0KiB muxing overhead: 0.372356%
frame= 17 fps=0.0 q=-1.0 Lsize=      622KiB time=00:00:12.73 bitrate= 399.8
kb/s speed= 415x elapsed=0:00:00.03
```

```
C:\Users\Dell\Pictures\Saved Pictures\video>
```

🖥️ > Pictures > Saved Pictures > video Search video 🔍 📁 🔗 🗑️ ↕️ Sort ▾ ☰ View ▾ ⋮ 📄 Details				
Name	Date	Type	Size	Tag
📹 input.mp4	2/27/2021 3:16 PM	MP4 File	8,611 KB	
📹 output.mp4	4/9/2026 3:12 PM	MP4 File	622 KB	

5. Merge Videos

a) Using concat

Create file list.txt

file 'video1.mp4'

file 'video2.mp4'

ffmpeg -f concat -safe 0 -i list.txt -c copy output.mp4

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -f concat -safe 0 -i list
.txt -c copy output.mp4
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
  built with gcc 15.2.0 (crosstool-NG 1.28.0.23_185f348)
  configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-opencore-amrnb --enable-
liblame --enable-liblcevc --enable-amf --enable-libaom --enable-libar
ibb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkvaza
r --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enab
le-libopenjpeg --enable-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enabl
e-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband --
enable-schannel --enable-sdl2 --enable-libsndio --enable-libsoxr --enable-
libsrt --enable-libsvtav1 --enable-libtwolame --enable-libuavs3d --disable-l
ibdrm --enable-vaapi --enable-libvidstab --enable-libvenc --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs=-extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags=- --cc=x8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm -
extra-version=20260407
  libavutil      60. 29.100 / 60. 29.100
  libavcodec     62. 29.101 / 62. 29.101
  libavformat    62. 13.101 / 62. 13.101
  libavdevice    62.  4.100 / 62.  4.100
  libavfilter    11. 15.101 / 11. 15.101
  libswscale      9.  7.100 /  9.  7.100
  libswsample     6.  4.100 /  6.  4.100
[mov,mp4,m4a,3gp,3g2,mj2 @ 0000025b52cdf740] Auto-inserting h264_mp4toannexb
bitstream filter
```

```

Press [q] to stop, [?] for help
[mov,mp4,m4a,3gp,3g2,mj2 @ 0000025b52d5a480] Auto-inserting h264_mp4toannexb
bitstream filter
[mov,mp4,m4a,3gp,3g2,mj2 @ 0000025b52cdf940] Auto-inserting h264_mp4toannexb
bitstream filter
[out#0/mp4 @ 0000025b52d4b740] video:24627KiB audio:1182KiB subtitle:0KiB ot
her streams:0KiB global headers:0KiB muxing overhead: 0.104736%
frame= 915 fps=0.0 q=-1.0 Lsize= 25836KiB time=00:00:38.20 bitrate=5539.7
kbits/s speed= 163x elapsed=0:00:00.23

```

Name	Date	Type	Size
 list.txt	4/9/2026 3:26 PM	Text Document	1 KB
 output.mp4	4/9/2026 3:27 PM	MP4 File	25,837 KB
 video1.mp4	2/27/2021 3:16 PM	MP4 File	8,611 KB
 video2.mp4	2/27/2021 3:16 PM	MP4 File	8,611 KB
 video3.mp4	2/27/2021 3:16 PM	MP4 File	8,611 KB

b) Using filter_complex

```
ffmpeg -i v1.mp4 -i v2.mp4 -filter_complex "[0:v][1:v]concat=n=2:v=1:a=0" output.mp4
```

Output

```
C:\Users\ DELL\ Pictures\ Saved Pictures\ video>ffmpeg -i v1.mp4 -i v2.mp4 -filt
er_complex "[0:v][1:v]concat=n=2:v=1:a=0" output.mp4
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
built with gcc 15.2.0 (crosstool-NG 1.28.0.23_185f348)
configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-openc1 --enable-gmp --e
nable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
ibb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkvaza
ar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enab
le-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enabl
e-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband -
--enable-schannel --enable-sdl2 --enable-libsnappy --enable-libsoxr --enable-
libsrt --enable-libsvtav1 --enable-libtwolame --enable-libuavs3d --disable-l
ibdrm --enable-vaapi --enable-libvidstab --enable-libvenc --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs= --extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags= --cc=x8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm -
--extra-version=20260407
libavutil      60. 29.100 / 60. 29.100
libavcodec     62. 29.101 / 62. 29.101
libavformat    62. 13.101 / 62. 13.101
libavdevice    62.  4.100 / 62.  4.100
libavfilter    11. 15.101 / 11. 15.101
libswscale     9.  7.100 / 9.  7.100
libswresample  6.  4.100 / 6.  4.100
Input #0, mov,mp4,m4a,3gp,3g2,mj2, from 'v1.mp4':
```

```

frame= 375 fps= 27 q=28.0 size= 4864KiB time=00:00:15.55 bitrate=2561.3k
frame= 387 fps= 27 q=28.0 size= 4864KiB time=00:00:16.05 bitrate=2481.4k
frame= 406 fps= 27 q=28.0 size= 4864KiB time=00:00:16.85 bitrate=2364.7k
frame= 419 fps= 27 q=28.0 size= 4864KiB time=00:00:17.39 bitrate=2291.0k
frame= 435 fps= 27 q=28.0 size= 4864KiB time=00:00:17.97 bitrate=2216.6k
frame= 453 fps= 27 q=28.0 size= 4864KiB time=00:00:18.72 bitrate=2127.7k
frame= 469 fps= 27 q=28.0 size= 4864KiB time=00:00:19.47 bitrate=2045.7k
frame= 485 fps= 28 q=28.0 size= 4864KiB time=00:00:20.14 bitrate=1978.0k
frame= 500 fps= 28 q=28.0 size= 4864KiB time=00:00:20.77 bitrate=1918.4k
frame= 517 fps= 28 q=28.0 size= 4864KiB time=00:00:21.47 bitrate=1855.1k
frame= 537 fps= 28 q=28.0 size= 4864KiB time=00:00:22.31 bitrate=1785.7k
frame= 561 fps= 29 q=28.0 size= 5120KiB time=N/A bitrate=N/A speed=N/A e
frame= 585 fps= 29 q=28.0 size= 5632KiB time=N/A bitrate=N/A speed=N/A e
[out#0/mp4 @ 0000016610fa92c0] video:10050KiB audio:4KiB subtitle:0KiB other
streams:0KiB global headers:0KiB muxing overhead: 0.135763%
frame= 610 fps= 29 q=-1.0 Lsize= 10067KiB time=00:00:25.35 bitrate=3252.0
kbits/s speed=1.23x elapsed=0:00:20.69
[libx264 @ 00000166110274c0] frame I:5 Avg QP:17.44 size:108277
[libx264 @ 00000166110274c0] frame P:155 Avg QP:20.96 size: 38864
[libx264 @ 00000166110274c0] frame B:450 Avg QP:23.54 size: 8277
[libx264 @ 00000166110274c0] consecutive B-frames: 1.1% 0.7% 2.5% 95.7%
[libx264 @ 00000166110274c0] mb I I16..4: 33.4% 52.8% 13.8%
[libx264 @ 00000166110274c0] mb P I16..4: 6.1% 7.4% 0.6% P16..4: 50.1%
13.7% 7.0% 0.0% 0.0% skip:15.1%
[libx264 @ 00000166110274c0] mb B I16..4: 0.8% 0.7% 0.0% B16..8: 41.5%
2.0% 0.2% direct: 2.2% skip:52.4% L0:41.4% L1:54.9% BI: 3.7%
[libx264 @ 00000166110274c0] 8x8 transform intra:51.5% inter:80.3%
[libx264 @ 00000166110274c0] coded y,uvDC,uvAC intra: 25.7% 69.6% 18.2% inte
r: 7.9% 22.8% 1.1%
[libx264 @ 00000166110274c0] i16 v,h,dc,p: 34% 24% 24% 18%
[libx264 @ 00000166110274c0] i8 v,h,dc,ddl,ddr,vr,hd,vl,hu: 18% 16% 45% 3%
4% 4% 4% 3% 3%
[libx264 @ 00000166110274c0] i4 v,h,dc,ddl,ddr,vr,hd,vl,hu: 23% 20% 18% 5%
8% 10% 7% 4% 5%
[libx264 @ 00000166110274c0] i8c dc,h,v,p: 52% 22% 19% 8%
[libx264 @ 00000166110274c0] Weighted P-Frames: Y:0.6% UV:0.6%
[libx264 @ 00000166110274c0] ref P L0: 64.3% 10.1% 19.5% 6.1%
[libx264 @ 00000166110274c0] ref B L0: 89.1% 9.0% 1.9%
[libx264 @ 00000166110274c0] ref B L1: 95.5% 4.5%
[libx264 @ 00000166110274c0] kb/s:3235.62
[aac @ 0000016612f2cbc0] Qavg: 65536.000

```

Name	Date	Type	Size	Tag
 output.mp4	4/9/2026 3:31 PM	MP4 File	10,067 KB	
 v1.mp4	2/27/2021 3:16 PM	MP4 File	8,611 KB	
 v2.mp4	2/27/2021 3:16 PM	MP4 File	8,611 KB	

6. Overlay Image / Text

Image overlay

```
ffmpeg -i input.mp4 -i logo.png -filter_complex "overlay=10:10" output.mp4
```

Text with timestamp

```
ffmpeg -i input.mp4 -vf
```

```
"drawtext=text='%{pts\:hms}':x=10:y=10:fontsize=24:fontcolor=white" output.mp4
```

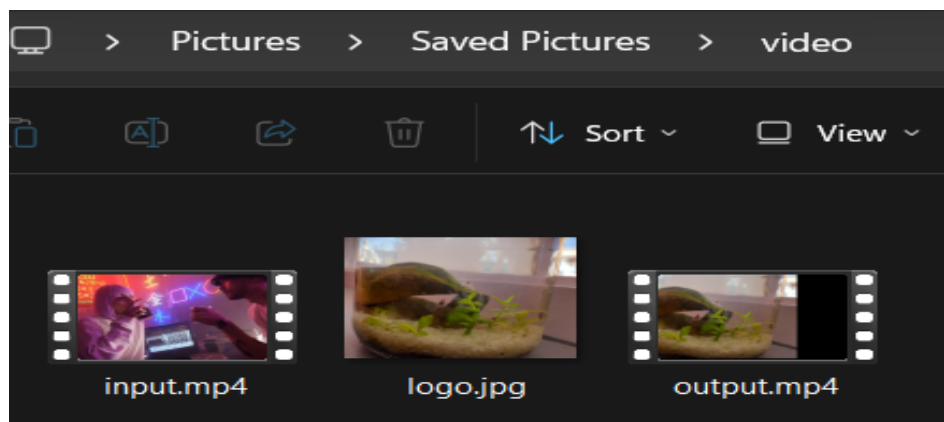
Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp4 -i logo.jpg
-filter_complex "overlay=10:10" output.mp4
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
  built with gcc 15.2.0 (crosstool-NG 1.28.0.23_185f348)
  configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-openc1 --enable-gmp --e
nable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
ibb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkvaza
ar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enab
le-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enabl
e-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband -
enable-schannel --enable-sdl2 --enable-libsndio --enable-libsoxr --enable-
libsrt --enable-libsvtav1 --enable-libtwolame --enable-libuavs3d --disable-l
ibdrm --enable-vaapi --enable-libvidstab --enable-libvenc --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs= --extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags= --cc=x8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm -
-extra-version=20260407
  libavutil      60. 29.100 / 60. 29.100
  libavcodec     62. 29.101 / 62. 29.101
  libavformat    62. 13.101 / 62. 13.101
  libavdevice    62.  4.100 / 62.  4.100
  libavfilter    11. 15.101 / 11. 15.101
  libswscale      9.  7.100 /  9.  7.100
  libswresample  6.  4.100 /  6.  4.100
Input #0, mov,mp4,m4a,3gp,3g2,mj2, from 'input.mp4':
Metadata:
```

```

frame= 49 fps= 19 q=28.0 size= 256KiB time=00:00:06.10 bitrate= 343.8k
frame= 65 fps= 20 q=28.0 size= 256KiB time=00:00:06.61 bitrate= 317.2k
frame= 85 fps= 23 q=28.0 size= 512KiB time=00:00:07.63 bitrate= 549.2k
frame= 108 fps= 25 q=28.0 size= 512KiB time=00:00:08.14 bitrate= 514.7k
frame= 123 fps= 25 q=28.0 size= 768KiB time=00:00:09.17 bitrate= 685.9k
frame= 149 fps= 28 q=28.0 size= 768KiB time=00:00:10.19 bitrate= 617.0k
frame= 175 fps= 29 q=28.0 size= 1024KiB time=00:00:11.22 bitrate= 747.6k
frame= 200 fps= 30 q=28.0 size= 1280KiB time=00:00:12.24 bitrate= 856.3k
frame= 225 fps= 31 q=28.0 size= 1280KiB time=00:00:09.30 bitrate=1127.4k
frame= 253 fps= 33 q=28.0 size= 1792KiB time=N/A bitrate=N/A speed=N/A e
frame= 282 fps= 34 q=28.0 size= 1792KiB time=N/A bitrate=N/A speed=N/A e
[out#0/mp4 @ 000001a5da6daa80] video:2082KiB audio:4KiB subtitle:0KiB other
streams:0KiB global headers:0KiB muxing overhead: 0.477261%
frame= 305 fps= 36 q=-1.0 Lsize= 2095KiB time=00:00:12.73 bitrate=1347.8
kbits/s speed=1.49x elapsed=0:00:08.53
[libx264 @ 000001a5dcd4f2c0] frame I:2 Avg QP:16.02 size:149109
[libx264 @ 000001a5dcd4f2c0] frame P:79 Avg QP:20.63 size: 13352
[libx264 @ 000001a5dcd4f2c0] frame B:224 Avg QP:22.33 size: 3474
[libx264 @ 000001a5dcd4f2c0] consecutive B-frames: 0.7% 0.7% 10.8% 87.9%
[libx264 @ 000001a5dcd4f2c0] mb I I16..4: 14.4% 70.8% 14.8%
[libx264 @ 000001a5dcd4f2c0] mb P I16..4: 3.8% 5.2% 0.4% P16..4: 21.1%
4.4% 2.1% 0.0% 0.0% skip:63.1%
[libx264 @ 000001a5dcd4f2c0] mb B I16..4: 0.4% 0.4% 0.0% B16..8: 16.1%
0.8% 0.1% direct: 1.3% skip:80.8% L0:40.5% L1:57.1% BI: 2.4%
[libx264 @ 000001a5dcd4f2c0] 8x8 transform intra:57.4% inter:88.3%
[libx264 @ 000001a5dcd4f2c0] coded y,uvDC,uvAC intra: 30.1% 66.1% 21.5% inte
r: 2.6% 9.0% 0.1%
[libx264 @ 000001a5dcd4f2c0] i16 v,h,dc,p: 33% 26% 24% 17%
[libx264 @ 000001a5dcd4f2c0] i8 v,h,dc,ddl,ddr,vr,hd,vl,hu: 21% 17% 42% 3%
4% 3% 4% 3% 3%
[libx264 @ 000001a5dcd4f2c0] i4 v,h,dc,ddl,ddr,vr,hd,vl,hu: 28% 24% 19% 4%
7% 5% 7% 3% 3%
[libx264 @ 000001a5dcd4f2c0] i8c dc,h,v,p: 51% 23% 20% 6%
[libx264 @ 000001a5dcd4f2c0] Weighted P-Frames: Y:0.0% UV:0.0%
[libx264 @ 000001a5dcd4f2c0] ref P L0: 67.6% 7.6% 18.1% 6.7%
[libx264 @ 000001a5dcd4f2c0] ref B L0: 88.8% 9.0% 2.2%
[libx264 @ 000001a5dcd4f2c0] ref B L1: 95.4% 4.6%
[libx264 @ 000001a5dcd4f2c0] kb/s:1340.27
[aac @ 000001a5dc850b00] Qavg: 65536.000

```



Name	Date	Type	Size	Length
 input.mp4	2/27/2021 3:16 PM	MP4 File	8,611 KB	00:00:12
 logo.jpg	12/8/2023 3:43 PM	JPG File	89 KB	
 output.mp4	4/9/2026 3:37 PM	MP4 File	2,096 KB	00:00:12

7. Filters

Greyscale

```
ffmpeg -i input.mp4 -vf format=gray output.mp4
```

Brightness/contrast

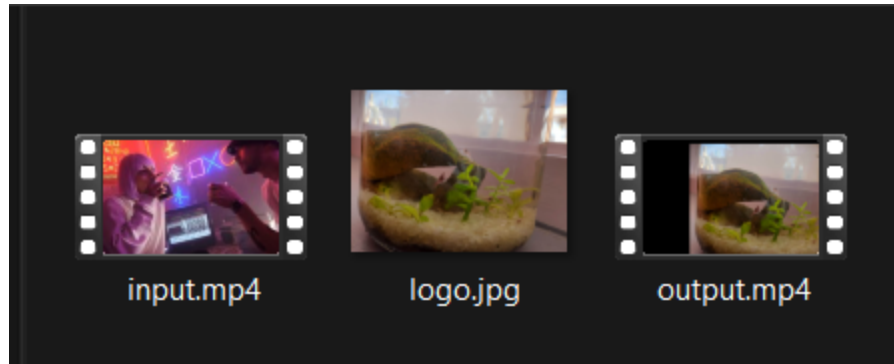
```
ffmpeg -i input.mp4 -vf eq=brightness=0.05:contrast=1.5 output.mp4
```

Watermark

```
ffmpeg -i input.mp4 -i logo.png -filter_complex "overlay=W-w-10:H-h-10" output.mp4
```

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp4 -i logo.jpg
-filter_complex "overlay=W-w-10:H-h-10" output.mp4
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFmpeg developers
  built with gcc 15.2.0 (cross-tool-NG 1.28.0.23_185f348)
  configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threads --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enable-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disable-libxcb --disable-xlib --disable-libpulse --enable-opencore-amrnb --enable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libaribb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libdav1d --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ffnvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkvazaar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libbrotli --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist --enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enable-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enable-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband --enable-schannel --enable-sdl2 --enable-libsndio --enable-libsoxr --enable-libsrt --enable-libsvtav1 --enable-libtwolame --enable-libuavs3d --disable-libdrm --enable-vaapi --enable-libvidstab --enable-libvvc --enable-whisper --enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enable-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxflags= --extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags= --cc=x86_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gcc-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm --extra-version=20260407
  libavutil      60. 29.100 / 60. 29.100
  libavcodec     62. 29.101 / 62. 29.101
  libavformat    62. 13.101 / 62. 13.101
  libavdevice    62.  4.100 / 62.  4.100
  libavfilter    11. 15.101 / 11. 15.101
  libswscale     9.  7.100 /  9.  7.100
  libswresample  6.  4.100 /  6.  4.100
Input #0, mov,mp4,m4a,3gp,3g2,mj2, from 'input.mp4':
  Metadata:
    major_brand      : mp42
```



8. Extract Audio / Video

Extract audio

```
ffmpeg -i input.mp4 -q:a 0 -map a output.mp3
```

Remove audio (video only)

```
ffmpeg -i input.mp4 -an output.mp4
```

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp4 -q:a 0 -map
a output.mp3
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
  built with gcc 15.2.0 (crosstool-NG 1.28.0.23_185f348)
  configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-openc1 --enable-gmp --e
nable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
ibb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkvaza
ar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enab
le-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enabl
e-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband -
--enable-schannel --enable-sdl2 --enable-libsnappy --enable-libsoxr --enable-
libsrt --enable-libsvtav1 --enable-libtwolame --enable-libuavs3d --disable-l
ibdrm --enable-vaapi --enable-libvidstab --enable-libvenc --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs= --extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags= --cc=x8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm -
extra-version=20260407
  libavutil      60. 29.100 / 60. 29.100
  libavcodec     62. 29.101 / 62. 29.101
  libavformat    62. 13.101 / 62. 13.101
  libavdevice    62.  4.100 / 62.  4.100
  libavfilter    11. 15.101 / 11. 15.101
  libswscale      9.  7.100 /  9.  7.100
  libswresample  6.  4.100 /  6.  4.100
Input #0, mov,mp4,m4a,3gp,3g2,mj2, from 'input.mp4':
```

```
  handler_name    : L-SMASH Audio Handler
[out#0/mp3 @ 0000027e05b88c00] video:0KiB audio:50KiB subtitle:0KiB other st
reams:0KiB global headers:0KiB muxing overhead: 0.661811%
size=      50KiB time=00:00:12.73 bitrate=  32.3kbits/s speed=97.5x elapsed=
0:00:00.13
```

```
C:\Users\Dell\Pictures\Saved Pictures\video>
```

 input.mp4	2/27/2021 3:16 PM	MP4 File	8,611 KB	00:00:12
 output.mp3	4/9/2026 3:44 PM	MP3 File	51 KB	00:00:12

9. Combine Audio & Video

ffmpeg -i video.mp4 -i audio.mp3 -c:v copy -c:a aac output.mp4

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp4 -i audio.mp3
-map 0:v -map 1:a -c:v copy -c:a aac output.mp4
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
```

```
built with gcc 15.2.0 (crosstool-NG 1.28.0.23_185f348)
configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-openc1 --enable-gmp --e
nable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
ibb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkvaza
r --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enab
le-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enabl
e-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband --
enable-schannel --enable-sdl2 --enable-lbsnappy --enable-libsoxr --enable-
libsrt --enable-libsvtav1 --enable-libttml --enable-libuavs3d --disable-l
ibdrm --enable-vaapi --enable-libvidstab --enable-libvenc --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-libzimg --enable-libzvb1 --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs= --extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags= --cc=x8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm -
extra-version=20260407
libavutil      60. 29.100 / 60. 29.100
libavcodec     62. 29.101 / 62. 29.101
libavformat    62. 13.101 / 62. 13.101
libavdevice    62.  4.100 / 62.  4.100
libavfilter    11. 15.101 / 11. 15.101
libswscale      9.  7.100 /  9.  7.100
libswresample  6.  4.100 /  6.  4.100
```

```
encoder       : Lavc62.29.101 aac
Stream #0:1: Audio: aac (LC) (mp4a / 0x6134706D), 48000 Hz, stereo, fltp,
128 kb/s
Metadata:
encoder       : Lavc62.29.101 aac
[out#0/mp4 @ 000002328496a980] video:8209KiB audio:4KiB subtitle:0KiB other
streams:0KiB global headers:0KiB muxing overhead: 0.120851%
frame= 305 fps=0.0 q=-1.0 Lsize= 8222KiB time=00:00:12.73 bitrate=5288.8
kb/s speed= 125x elapsed=0:00:00.10
[aac @ 0000023284980e80] Qavg: 65536.000
```

Name	Date	Type	Size	Length
 audio.mp3	4/9/2026 4:11 PM	MP3 File	51 KB	00:00:12
 input.mp4	2/27/2021 3:16 PM	MP4 File	8,611 KB	00:00:12
 output.mp4	4/9/2026 4:12 PM	MP4 File	8,223 KB	00:00:12

```
ffmpeg -framerate 1/3 -i img%d.jpg -c:v libx264 -r 30 output.mp4
```

[illegible]

```

frame= 316 fps=5.3 q=29.0 size= 4352KiB time=00:00:10.46 bitrate=3406.2k
frame= 324 fps=5.4 q=29.0 size= 4352KiB time=00:00:10.73 bitrate=3321.6k
frame= 340 fps=5.6 q=-1.0 size= 4352KiB time=00:00:11.26 bitrate=3164.4k
frame= 344 fps=5.6 q=-1.0 size= 4352KiB time=00:00:11.40 bitrate=3127.4k
[out#0/mp4 @ 000001c322ce9340] video:4450KiB audio:0KiB subtitle:0KiB other
streams:0KiB global headers:0KiB muxing overhead: 0.183151%
frame= 360 fps=5.8 q=-1.0 Lsize= 4458KiB time=00:00:11.93 bitrate=3060.5
kbits/s dup=356 drop=0 speed=0.194x elapsed=0:01:01.54
[libx264 @ 000001c324c87ac0] frame I:4 Avg QP:15.00 size:1095588
[libx264 @ 000001c324c87ac0] frame P:92 Avg QP:18.13 size: 709
[libx264 @ 000001c324c87ac0] frame B:264 Avg QP:22.86 size: 412
[libx264 @ 000001c324c87ac0] consecutive B-frames: 2.2% 0.0% 0.0% 97.8%
[libx264 @ 000001c324c87ac0] mb I I16..4: 3.7% 84.8% 11.5%
[libx264 @ 000001c324c87ac0] mb P I16..4: 0.0% 0.0% 0.0% P16..4: 0.5%
0.0% 0.0% 0.0% 0.0% skip:99.5%
[libx264 @ 000001c324c87ac0] mb B I16..4: 0.0% 0.0% 0.0% B16..8: 0.2%
0.0% 0.0% direct: 0.0% skip:99.8% L0:46.0% L1:54.0% BI: 0.0%
[libx264 @ 000001c324c87ac0] 8x8 transform intra:84.8% inter:95.9%
[libx264 @ 000001c324c87ac0] coded y,uvDC,uvAC intra: 87.7% 64.0% 39.2% inte
r: 0.0% 0.1% 0.0%
[libx264 @ 000001c324c87ac0] i16 v,h,dc,p: 39% 21% 28% 11%
[libx264 @ 000001c324c87ac0] i8 v,h,dc,ddl,ddr,vr,hd,vl,hu: 23% 21% 28% 4%
4% 5% 4% 5% 6%
[libx264 @ 000001c324c87ac0] i4 v,h,dc,ddl,ddr,vr,hd,vl,hu: 46% 17% 6% 4%
6% 7% 4% 6% 3%
[libx264 @ 000001c324c87ac0] i8c dc,h,v,p: 39% 23% 31% 7%
[libx264 @ 000001c324c87ac0] Weighted P-Frames: Y:0.0% UV:0.0%
[libx264 @ 000001c324c87ac0] ref P L0: 93.6% 0.1% 2.7% 3.7%
[libx264 @ 000001c324c87ac0] ref B L0: 86.9% 6.0% 7.1%
[libx264 @ 000001c324c87ac0] ref B L1: 99.1% 0.9%
[libx264 @ 000001c324c87ac0] kb/s:3037.47

```

 img1.jpg	12/21/2023 9:25 PM	JPG File	575 KB	
 img2.jpg	12/21/2023 9:25 PM	JPG File	444 KB	
 img3.jpg	12/21/2023 9:25 PM	JPG File	601 KB	
 img4.jpg	12/21/2023 9:25 PM	JPG File	654 KB	
 input.mp4	2/27/2021 3:16 PM	MP4 File	8,611 KB	00:00:12
 output.mp4	4/9/2026 4:18 PM	MP4 File	4,459 KB	00:00:12

11. Add Subtitles

```
ffmpeg -i input.mp4 -vf subtitles=sub.srt output.mp4
```

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp4 -vf subtitle
s=sub.srt output.mp4
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
  built with gcc 15.2.0 (crosstool-NG 1.28.0.23_185f348)
  configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-openc1 --enable-gmp --e
nable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
ibb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkvaza
ar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enab
le-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enabl
e-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband -
--enable-schannel --enable-sdl2 --enable-libsndio --enable-libsoxr --enable-
libsrt --enable-libsvtav1 --enable-libtwolame --enable-libuavs3d --disable-l
ibdrm --enable-vaapi --enable-libvidstab --enable-libvenc --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs= --extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags= --cc=x8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm -
--extra-version=20260407
  libavutil      60. 29.100 / 60. 29.100
  libavcodec     62. 29.101 / 62. 29.101
  libavformat    62. 13.101 / 62. 13.101
  libavdevice    62.  4.100 / 62.  4.100
  libavfilter    11. 15.101 / 11. 15.101
  libswscale      9.  7.100 /  9.  7.100
  libswresample  6.  4.100 /  6.  4.100
```

```

frame= 227 fps= 29 q=28.0 size=    3584KiB time=00:00:09.38 bitrate=3128.7k
frame= 253 fps= 30 q=28.0 size=    4096KiB time=N/A bitrate=N/A speed=N/A e
frame= 279 fps= 31 q=28.0 size=    4608KiB time=N/A bitrate=N/A speed=N/A e
[out#0/mp4 @ 000001c0a088cdc0] video:5008KiB audio:4KiB subtitle:0KiB other
streams:0KiB global headers:0KiB muxing overhead: 0.200329%
frame= 305 fps= 33 q=-1.0 lsize=    5022KiB time=00:00:12.73 bitrate=3230.1
kb/s speed=1.36x elapsed=0:00:09.34
[libx264 @ 000001c0a08a2240] frame I:2      Avg QP:16.13  size: 79691
[libx264 @ 000001c0a08a2240] frame P:77     Avg QP:21.07  size: 39949
[libx264 @ 000001c0a08a2240] frame B:226    Avg QP:23.65  size: 8373
[libx264 @ 000001c0a08a2240] consecutive B-frames:  0.7%  1.3%  1.0% 97.0%
[libx264 @ 000001c0a08a2240] mb I  I16..4: 54.5% 34.1% 11.4%
[libx264 @ 000001c0a08a2240] mb P  I16..4:  6.1%  8.1%  0.7% P16..4: 49.7%
13.7%  7.2%  0.0%  0.0%  skip:14.5%
[libx264 @ 000001c0a08a2240] mb B  I16..4:  0.8%  0.7%  0.0% B16..8: 41.8%
 2.1%  0.2% direct: 2.3% skip:52.1% L0:42.0% L1:54.5% BI: 3.6%
[libx264 @ 000001c0a08a2240] 8x8 transform intra:50.2% inter:80.8%
[libx264 @ 000001c0a08a2240] coded y,uvDC,uvAC intra: 24.3% 67.1% 17.1% inte
r: 8.1% 23.0% 1.0%
[libx264 @ 000001c0a08a2240] i16 v,h,dc,p: 38% 23% 22% 17%
[libx264 @ 000001c0a08a2240] i8  v,h,dc,ddl,ddr,vr,hd,vl,hu: 19% 16% 45%  3%
 4%  4%  4%  3%  3%
[libx264 @ 000001c0a08a2240] i4  v,h,dc,ddl,ddr,vr,hd,vl,hu: 24% 20% 17%  5%
 8%  9%  7%  5%  5%
[libx264 @ 000001c0a08a2240] i8c dc,h,v,p: 53% 22% 18%  7%
[libx264 @ 000001c0a08a2240] Weighted P-Frames: Y:0.0% UV:0.0%
[libx264 @ 000001c0a08a2240] ref P L0: 63.7% 10.3% 20.0%  5.9%
[libx264 @ 000001c0a08a2240] ref B L0: 89.8%  8.5%  1.8%
[libx264 @ 000001c0a08a2240] ref B L1: 95.4%  4.6%
[libx264 @ 000001c0a08a2240] kb/s:3224.74
[aac @ 000001c0a2988600] Qavg: 65536.000

```

 input.mp4	2/27/2021 3:16 PM	MP4 File	8,611 KB	00:00:12
 output.mp4	4/9/2026 4:24 PM	MP4 File	5,022 KB	00:00:12
 sub.srt	4/9/2026 4:24 PM	SRT File	1 KB	

12. Create GIF

```
ffmpeg -i input.mp4 -vf "fps=10,scale=320:-1" output.gif
```

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp4 -vf "fps=10,
scale=320:-1" output.gif
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
  built with gcc 15.2.0 (crosstool-NG 1.28.0.23_185f348)
  configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-openc1 --enable-gmp --e
nable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
ibb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkvaza
ar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enab
le-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enabl
e-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband -
--enable-schannel --enable-sdl2 --enable-libsndio --enable-libsoxr --enable-
libsrt --enable-libsvtav1 --enable-libttml --enable-libuavs3d --disable-l
ibdrm --enable-vaapi --enable-libvidstab --enable-libvvc --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs= --extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags= --cc=x8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm -
--extra-version=20260407
  libavutil      60. 29.100 / 60. 29.100
  libavcodec     62. 29.101 / 62. 29.101
  libavformat    62. 13.101 / 62. 13.101
  libavdevice    62.  4.100 / 62.  4.100
  libavfilter    11. 15.101 / 11. 15.101
  libswscale      9.  7.100 /  9.  7.100
  libswresample  6.  4.100 /  6.  4.100
  handler_name    : E SHASH Video Handler
frame=   29 fps=0.0 q=-0.0 size=    256KiB time=00:00:02.90 bitrate= 723.2k
frame=   78 fps= 76 q=-0.0 size=   1024KiB time=00:00:07.80 bitrate=1075.5k
frame=  123 fps= 79 q=-0.0 size=   1792KiB time=00:00:12.30 bitrate=1193.5k
[out#0/gif @ 000001b58be4c1c0] video:1986KiB audio:0KiB subtitle:0KiB other
streams:0KiB global headers:0KiB muxing overhead: 0.000984%
frame=  127 fps= 80 q=-0.0 Lsize=   1986KiB time=00:00:12.70 bitrate=1280.8
kbits/s speed=7.99x elapsed=0:00:01.58
```

 input.mp4	2/27/2021 3:16 PM	MP4 File	8,611 KB	00:00:12
 output.gif	4/9/2026 4:27 PM	GIF File	1,986 KB	

13. Extract Frames

```
ffmpeg -i input.mp4 frame_%04d.png
```

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp4 frame_%04d.p
ng
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
  built with gcc 15.2.0 (crosstool-NG 1.28.0.23_185f348)
  configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-openccl --enable-gmp --e
nable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
ibb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkvaza
r --enable-libribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable-
libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enab
le-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enabl
e-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband -
--enable-schannel --enable-sdl2 --enable-libsnpappy --enable-libsoxr --enable-
libsrt --enable-libsvtav1 --enable-libtwolame --enable-libuavs3d --disable-l
ibdrm --enable-vaapi --enable-libvidstab --enable-libvenc --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs= --extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags= --cc=x8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm -
-extra-version=20260407
  libavutil      60. 29.100 / 60. 29.100
  libavcodec     62. 29.101 / 62. 29.101
  libavformat    62. 13.101 / 62. 13.101
  libavdevice    62.  4.100 / 62.  4.100
  libavfilter    11. 15.101 / 11. 15.101
  libswscale      9.  7.100 /  9.  7.100
  libswresample  6.  4.100 /  6.  4.100
```

```

frame= 252 fps= 11 q=-0.0 size=N/A time=00:00:10.51 bitrate=N/A speed=0.479
frame= 260 fps= 12 q=-0.0 size=N/A time=00:00:10.84 bitrate=N/A speed=0.483
frame= 269 fps= 12 q=-0.0 size=N/A time=00:00:11.21 bitrate=N/A speed=0.487
frame= 277 fps= 12 q=-0.0 size=N/A time=00:00:11.55 bitrate=N/A speed=0.491
frame= 281 fps= 12 q=-0.0 size=N/A time=00:00:11.72 bitrate=N/A speed=0.486
frame= 291 fps= 12 q=-0.0 size=N/A time=00:00:12.13 bitrate=N/A speed=0.492
frame= 298 fps= 12 q=-0.0 size=N/A time=00:00:12.42 bitrate=N/A speed=0.493
[out#0/image2 @ 000002bc8f8f9400] video:483808KiB audio:0KiB subtitle:0KiB o
ther streams:0KiB global headers:0KiB muxing overhead: unknown
frame= 305 fps= 12 q=-0.0 Lsize=N/A time=00:00:12.72 bitrate=N/A speed= 0.5
x elapsed=0:00:25.46

```

Name	Date	Type	Size
 frame_0001.png	4/9/2026 4:30 PM	PNG File	9 KB
 frame_0002.png	4/9/2026 4:30 PM	PNG File	1,379 KB
 frame_0003.png	4/9/2026 4:30 PM	PNG File	1,487 KB
 frame_0004.png	4/9/2026 4:30 PM	PNG File	1,504 KB
 frame_0005.png	4/9/2026 4:30 PM	PNG File	1,516 KB
 frame_0006.png	4/9/2026 4:30 PM	PNG File	1,535 KB
 frame_0007.png	4/9/2026 4:30 PM	PNG File	1,544 KB
 frame_0285.png	4/9/2026 4:31 PM	PNG File	1,569 KB
 frame_0286.png	4/9/2026 4:31 PM	PNG File	1,560 KB
 frame_0287.png	4/9/2026 4:31 PM	PNG File	1,569 KB
 frame_0288.png	4/9/2026 4:31 PM	PNG File	1,592 KB
 frame_0289.png	4/9/2026 4:31 PM	PNG File	1,499 KB
 frame_0290.png	4/9/2026 4:31 PM	PNG File	1,538 KB
 frame_0291.png	4/9/2026 4:31 PM	PNG File	1,545 KB
 frame_0292.png	4/9/2026 4:31 PM	PNG File	1,551 KB
 frame_0293.png	4/9/2026 4:31 PM	PNG File	1,554 KB
 frame_0294.png	4/9/2026 4:31 PM	PNG File	1,553 KB
 frame_0295.png	4/9/2026 4:31 PM	PNG File	1,562 KB
 frame_0296.png	4/9/2026 4:31 PM	PNG File	1,567 KB
 frame_0297.png	4/9/2026 4:31 PM	PNG File	1,571 KB
 frame_0298.png	4/9/2026 4:31 PM	PNG File	1,567 KB
 frame_0299.png	4/9/2026 4:31 PM	PNG File	1,576 KB
 frame_0300.png	4/9/2026 4:31 PM	PNG File	1,581 KB
 frame_0301.png	4/9/2026 4:31 PM	PNG File	1,610 KB
 frame_0302.png	4/9/2026 4:31 PM	PNG File	1,614 KB
 frame_0303.png	4/9/2026 4:31 PM	PNG File	1,604 KB
 frame_0304.png	4/9/2026 4:31 PM	PNG File	1,586 KB
 frame_0305.png	4/9/2026 4:31 PM	PNG File	1,560 KB
 input.mp4	2/27/2021 3:16 PM	MP4 File	8,611 KB 00:00:12

```
ffmpeg -i input.mp4 -filter:v "setpts=2.0*PTS" output.mp4
```

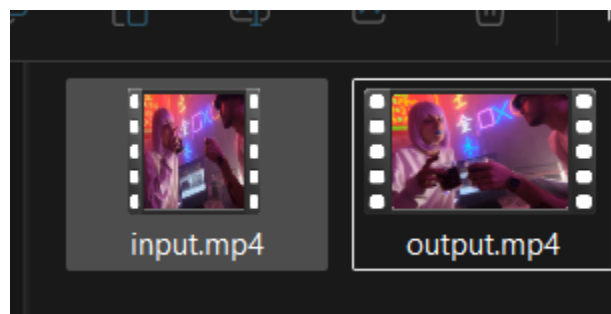
```
C:\Users\ DELL\Pictures\Saved Pictures\video>ffmpeg -i input.mp4 -filter:v "s
etpts=2.0*PTS" output.mp4
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
  built with gcc 15.2.0 (crosstool-NG 1.28.0.23_185f348)
  configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-openc1 --enable-gmp --e
nable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
ib24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkvaza
ar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enab
le-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enabl
e-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband -
-enable-schannel --enable-sdl2 --enable-libsndio --enable-libsoxr --enable-
libsrt --enable-libsvtav1 --enable-libttml --enable-libuavs3d --disable-l
ibdrm --enable-vaapi --enable-libvidstab --enable-libvenc --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs= --extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags= --cc=x8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm -
-extra-version=20260407
  libavutil      60. 29.100 / 60. 29.100
  libavcodec     62. 29.101 / 62. 29.101
  libavformat    62. 13.101 / 62. 13.101
  libavdevice    62.  4.100 / 62.  4.100
  libavfilter    11. 15.101 / 11. 15.101
  libswscale      9.  7.100 /  9.  7.100
  libswresample  6.  4.100 /  6.  4.100
Input #0, mov,mp4,m4a,3gp,3g2,mj2, from 'input.mp4':
Metadata:
  major_brand      : mp42
```

```

frame= 227 fps= 26 q=26.0 size=    3584KiB time=00:00:18.72 bitrate=1567.8k
frame= 252 fps= 27 q=26.0 size=    3584KiB time=N/A bitrate=N/A speed=N/A e
frame= 272 fps= 28 q=26.0 size=    3584KiB time=N/A bitrate=N/A speed=N/A e
frame= 287 fps= 28 q=-1.0 size=    3840KiB time=N/A bitrate=N/A speed=N/A e
[out#0/mp4 @ 000002615fa594c0] video:7050KiB audio:4KiB subtitle:0KiB other
streams:0KiB global headers:0KiB muxing overhead: 0.126508%
frame= 305 fps= 29 q=-1.0 Lsize=    7062KiB time=00:00:25.23 bitrate=2292.7
kbits/s speed=2.37x elapsed=0:00:10.64
[libx264 @ 0000026161eb5400] frame I:3      Avg QP:14.94  size:105262
[libx264 @ 0000026161eb5400] frame P:77     Avg QP:18.68  size: 54281
[libx264 @ 0000026161eb5400] frame B:225    Avg QP:21.41  size: 12101
[libx264 @ 0000026161eb5400] consecutive B-frames:  1.0%  1.3%  2.0% 95.7%
[libx264 @ 0000026161eb5400] mb I  I16..4: 43.0% 43.7% 13.3%
[libx264 @ 0000026161eb5400] mb P  I16..4:  6.5%  9.1%  1.1% P16..4: 47.9%
18.8%  8.9%  0.0%  0.0%  skip: 7.8%
[libx264 @ 0000026161eb5400] mb B  I16..4:  1.2%  1.1%  0.0% B16..8: 45.9%
 3.7%  0.4% direct: 3.7% skip:43.9% L0:42.4% L1:51.8% BI: 5.8%
[libx264 @ 0000026161eb5400] 8x8 transform intra:51.2% inter:76.0%
[libx264 @ 0000026161eb5400] coded y,uvDC,uvAC intra: 29.6% 71.8% 24.4% inte
r: 11.5% 29.5% 1.7%
[libx264 @ 0000026161eb5400] i16 v,h,dc,p: 36% 22% 24% 17%
[libx264 @ 0000026161eb5400] i8  v,h,dc,ddl,ddr,vr,hd,vl,hu: 20% 17% 42%  3%
 4% 4% 4% 3% 3%
[libx264 @ 0000026161eb5400] i4  v,h,dc,ddl,ddr,vr,hd,vl,hu: 26% 21% 22%  4%
 7% 8% 6% 4% 4%
[libx264 @ 0000026161eb5400] i8c dc,h,v,p: 51% 22% 19%  9%
[libx264 @ 0000026161eb5400] Weighted P-Frames: Y:0.0% UV:0.0%
[libx264 @ 0000026161eb5400] ref P L0: 63.4% 10.1% 19.8%  6.8%
[libx264 @ 0000026161eb5400] ref B L0: 87.8% 10.2%  2.1%
[libx264 @ 0000026161eb5400] ref B L1: 94.9%  5.1%
[libx264 @ 0000026161eb5400] kb/s:2269.68
[aac @ 0000026161b524c0] Qavg: 65536.000

```

 input.mp4	2/27/2021 3:16 PM	MP4 File	8,611 KB	00:00:12
 output.mp4	4/9/2026 4:34 PM	MP4 File	7,063 KB	00:00:25



Audio Tasks

1. Conversion

ffmpeg -i input.mp3 output.wav

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp3 output.wav
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFmpeg developers
  built with gcc 15.2.0 (crosstool-NG 1.28.0.23_185f348)
  configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threads --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enable-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disable-libxcb --disable-xlib --disable-libpulse --enable-opencore-amrnb --enable-gmp --enable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libaribb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libdavs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ffnvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkvazaar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libbrotli --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist --enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enable-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enable-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband --enable-schannel --enable-sdl2 --enable-libsndio --enable-libsoxr --enable-libsrtp --enable-libsvtav1 --enable-libtwolame --enable-libuavs3d --disable-libdrm --enable-vaapi --enable-libvidstab --enable-libvvaenc --enable-whisper --enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enable-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxflags= --extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags= --cc=x86_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gcc-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm --extra-version=20260407

encoder          : Lavc62.29.101 pcm_s16le
[out#0/wav @ 000001b9a2efb540] video:0KiB audio:2388KiB subtitle:0KiB other streams:0KiB global headers:0KiB muxing overhead: 0.003190%
size= 2388KiB time=00:00:12.73 bitrate=1536.0kbits/s speed= 317x elapsed=0:00:00.04
```

 input.mp3	4/9/2026 8:47 PM	MP3 File	51 KB	00:00:12
 input.mp4	2/27/2021 3:16 PM	MP4 File	8,611 KB	00:00:12
 output.wav	4/9/2026 8:47 PM	WAV File	2,389 KB	00:00:12

2. Adjust Audio Quality

Bitrate

ffmpeg -i input.mp3 -b:a 192k output.mp3

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp3 -b:a 192k ou
tput.mp3
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
  built with gcc 15.2.0 (crosstool-NG 1.28.0.23_185f348)
  configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-openc1 --enable-gmp --e
nable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
ibb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkvaza
r --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enab
le-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enabl
e-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband -
--enable-schannel --enable-sdl2 --enable-libsnpappy --enable-libsoxr --enable-
libsrt --enable-libsvtav1 --enable-libtwolame --enable-libuavs3d --disable-l
ibdrm --enable-vaapi --enable-libvidstab --enable-libvenc --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs= --extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags= --cc=x8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm -
extra-version=20260407
  libavutil      60. 29.100 / 60. 29.100
  libavcodec     62. 29.101 / 62. 29.101
  libavformat    62. 13.101 / 62. 13.101
  libavdevice    62.  4.100 / 62.  4.100
  libavfilter    11. 15.101 / 11. 15.101
  libswscale      9.  7.100 /  9.  7.100
  libswresample  6.  4.100 /  6.  4.100
  encoder:       : Lame02:29:101 libmp3lame
[out#0/mp3 @ 0000016ad41acc00] video:0KiB audio:299KiB subtitle:0KiB other s
treams:0KiB global headers:0KiB muxing overhead: 0.235615%
size= 300KiB time=00:00:12.73 bitrate= 192.9kbits/s speed=74.4x elapsed=
0:00:00.17

C:\Users\Dell\Pictures\Saved Pictures\video>|
```

 input.mp3	4/9/2026 8:47 PM	MP3 File	51 KB	00:00:12
 output.mp3	4/9/2026 8:53 PM	MP3 File	300 KB	00:00:12

```
ffmpeg -i input.mp3 -ar 44100 output.mp3
```

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp3 -ar 44100 ou
tput.mp3
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
built with gcc 15.2.0 (crosstool-NG 1.28.0.23_185f348)
configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-opencl --enable-gmp --e
nable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
ibb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkva
ar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-openssl --enable-liboapv --enab
le-libopencl --enable-libopencl-amrnb --enable-libopencl-amrwb --enable-libopenh264 --enabl
e-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband -
--enable-schannel --enable-sdl2 --enable-libsndio --enable-libsoxr --enable-
libsrt --enable-libsvtav1 --enable-libtwolame --enable-libuavs3d --disable-l
ibdrm --enable-vaapi --enable-libvidstab --enable-libvvc --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs= --extra-libs=lgomp --extra-ldflags=-pthread --extra-ldexeflags= --cc=x8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm -
--extra-version=20260407
```

```
encoder      : Lavc02.29.101 libmp3lame
[out#0/mp3 @ 000001818d83adc0] video:0KiB audio:200KiB subtitle:0KiB other s
streams:0KiB global headers:0KiB muxing overhead: 0.173205%
size=      200KiB time=00:00:12.73 bitrate= 128.6kbits/s speed=80.3x elapsed=
0:00:00.15
```

Name	Date	Type	Size	Length
 input.mp3	4/9/2026 8:47 PM	MP3 File	51 KB	00:00:12
 output.mp3	4/9/2026 9:03 PM	MP3 File	200 KB	00:00:13

3. Audio Filters

Normalize

```
ffmpeg -i input.mp3 -af loudnorm output.mp3
```

Output

```
C:\Users\Dell\Pictures\Saved Pictures>ffmpeg -i input.mp3 -af loudnorm  
output.mp3  
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFmpeg developers  
built with gcc 15.2.0 (cross-tool-NG 1.28.0.23_185f348)  
configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threads --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enable-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disable-libxcb --disable-xlib --disable-libpulse --enable-opencl --enable-gmp --enable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libaribbb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libdav1d --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ffvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkvazaar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjxl --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist --enable-libssh --enable-libtheora --enable-libvp8 --enable-libwebp --enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enable-libzimg --enable-libzvbi --extra-cflags=-DLIBTOWOLAME_STATIC --extra-cxxflags= --extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags=-ccx86_64-w64-mingw32-gcc --ccx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gcc-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm --extra-version=20260407
```

```
size=      57KiB time=00:00:03.60 bitrate= 128.7kbits/s speed=   7x elapsed=
size=     132KiB time=00:00:08.40 bitrate= 128.3kbits/s speed=8.13x elapsed=
[out#0/mp3 @ 00000234fbe0d700] video:0KiB audio:200KiB subtitle:0KiB other s
treams:0KiB global headers:0KiB muxing overhead: 0.165453%
size=     200KiB time=00:00:12.80 bitrate= 127.9kbits/s speed=9.09x elapsed=
0:00:01.40
```

Name	Date	Type	Size	Length
 input.mp3	4/9/2026 8:47 PM	MP3 File	51 KB	00:00:12
 output.mp3	4/9/2026 9:11 PM	MP3 File	200 KB	00:00:13

Volume increase

```
ffmpeg -i input.mp3 -af "volume=2.0" output.mp3
```

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp3 -af "volume=
2.0" output.mp3
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
  built with gcc 15.2.0 (crosstool-NG 1.28.0.23_185f348)
  configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-openc1 --enable-gmp --e
nable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
ibb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkva
za --enable-libaribcaption --enable-libass --enable-libbluray --enable-libj
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enab
le-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enabl
e-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband -
-enable-schannel --enable-sdl2 --enable-libsndio --enable-libsoxr --enable-
libsrt --enable-libsvtav1 --enable-libtwolame --enable-libuavs3d --disable-l
ibdrm --enable-vaapi --enable-libvidstab --enable-libvenc --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs= --extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags= --cc=x8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm -
-extra-version=20260407
```

```
encoder      : Lavc62.29.101 libmp3lame
[out#0/mp3 @ 000001f9668eb800] video:0KiB audio:200KiB subtitle:0KiB other s
treams:0KiB global headers:0KiB muxing overhead: 0.165453%
size=       200KiB time=00:00:12.73 bitrate= 128.5kbits/s speed= 101x elapsed=
0:00:00.12
```

```
C:\Users\Dell\Pictures\Saved Pictures\video>
```

Name	Date	Type	Size	Length
 input.mp3	4/9/2026 8:47 PM	MP3 File	51 KB	00:00:12
 output.mp3	4/9/2026 9:17 PM	MP3 File	200 KB	00:00:13

```
ffmpeg -i input.mp3 -af "highpass=f=200" output.mp3
```

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp3 -af "highpas
s=f=200" output.mp3
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
built with gcc 15.2.0 (crosstool-NG 1.28.0.23_185f348)
configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-opencl --enable-gmp --e
nable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
ibb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkvaza
ar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enab
le-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enabl
e-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband -
-enable-schannel --enable-sdl2 --enable-libsndio --enable-libsoxr --enable-
libsrt --enable-libsvtav1 --enable-libtwinlame --enable-libuavs3d --disable-l
ibdrm --enable-vaapi --enable-libvidstab --enable-libvvc --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs= --extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags= --cc=x8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm -
-extra-version=20260407
```

C:\Users\Dell\Pictures\Saved Pictures\video>

Name	Date	Type	Size	Length
 input.mp3	4/9/2026 8:47 PM	MP3 File	51 KB	00:00:12
 output.mp3	4/9/2026 9:20 PM	MP3 File	200 KB	00:00:13

```
ffmpeg -i input.mp3 -af "lowpass=f=3000" output.mp3
```

```
C:\Users\Dell\Pictures\Saved Pictures\video> ffmpeg -i input.mp3 -af "lowpass=f=3000" output.mp3
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFmpeg developers
built with gcc 15.2.0 (crosstool-NG 1.28.0.23_185f348)
configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threads --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enable-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disable-libxcb --disable-xlib --disable-libpulse --enable-openccl --enable-gmp --enable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libaribb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libdav1d --enable-libdvbav2 --enable-libdvbnav --disable-libfdk-aac --enable-ffnvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkvazaar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjxl --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist --enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enable-libopencore-amrn --enable-libopencore-amrwb --enable-libopenh264 --enable-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband --enable-schannel --enable-sdl2 --enable-libsnpappy --enable-libsoxr --enable-libsrt --enable-libsvtav1 --enable-libtwolame --enable-libuavs3d --disable-libdrm --enable-vaapi --enable-libvidstab --enable-libvvec --enable-whisper --enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enable-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxflags= --extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags= --cc=x86_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gcc-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm --extra-version=20260407
encoder         : Lavc62.29.101 libmp3lame
[out#0/mp3 @ 000002565221fbc0] video:0KiB audio:200KiB subtitle:0KiB other streams:0KiB global headers:0KiB muxing overhead: 0.165453%
size=          200KiB time=00:00:12.73 bitrate= 128.5kbits/s speed= 104x elapsed= 0:00:00.12

C:\Users\Dell\Pictures\Saved Pictures\video>
```

Name	Date	Type	Size	Length
 input.mp3	4/9/2026 8:47 PM	MP3 File	51 KB	00:00:12
 output.mp3	4/9/2026 9:25 PM	MP3 File	200 KB	00:00:13

```
ffmpeg -i input.mp3 -af "aecho=0.8:0.9:1000:0.3" output.mp3
```

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp3 -af "aecho=0
.8:0.9:1000:0.3" output.mp3
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
  built with gcc 15.2.0 (crosstool-NG 1.28.0.23_185f348)
  configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-openc1 --enable-gmp --e
nable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
ibb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkvaza
r --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enab
le-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enabl
e-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband -
--enable-schannel --enable-sdl2 --enable-libsnappy --enable-libsoxr --enable-
libsrt --enable-libsvtav1 --enable-libtwolame --enable-libuavs3d --disable-l
ibdrm --enable-vaapi --enable-libvidstab --enable-libvenc --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs= --extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags= --cc=x8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm -
--extra-version=20260407
  libavutil      60. 29.100 / 60. 29.100
```

```
encoder      : Lavc62.29.101 libmp3lame
[out#0/mp3 @ 0000015893f5a440] video:0KiB audio:215KiB subtitle:0KiB other streams:0KiB global headers:0KiB muxing overhead: 0.153346%
size=      216KiB time=00:00:13.73 bitrate= 128.6kbits/s speed= 109x elapsed=0:00:00.12
```

```
C:\Users\Dell\Pictures\Saved Pictures\video>
```

Name	Date	Type	Size	Length
 input.mp3	4/9/2026 8:47 PM	MP3 File	51 KB	00:00:12
 output.mp3	4/9/2026 9:26 PM	MP3 File	216 KB	00:00:14

Pitch change

ffmpeg -i input.mp3 -af "asetrate=44100*1.25" output.mp3

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp3 -af "asetrate=44100*1.25" output.mp3
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFmpeg developers
```

```
  built with gcc 15.2.0 (crosstool-NG 1.28.0.23_185f348)
```

```
  configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threads --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enable-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disable-libxcb --disable-xlib --disable-libpulse --enable-openccl --enable-gmp --enable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libaribb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libdav1d --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ffnvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkvazaar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libbrotli --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist --enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enable-libopencl --enable-libopenmpt --enable-librav1e --enable-librubberband --enable-schannel --enable-sdl2 --enable-libsrt --enable-libsoxr --enable-libsrt --enable-libsrt --enable-libswscale2 --enable-libtesseract --enable-libtheora --enable-libtwolame --enable-libuavs3d --disable-libdrm --enable-vaapi --enable-libvidstab --enable-libvvaenc --enable-whisper --enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enable-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxflags= --extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags= --cc=x86_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gcc-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm --extra-version=20260407
```

```
[out#0/mp3 @ 000001ae22fbfc80] video:0KiB audio:174KiB subtitle:0KiB other streams:0KiB global headers:0KiB muxing overhead: 0.189700%
size=      174KiB time=00:00:11.08 bitrate= 128.8kbits/s speed=94.2x elapsed=0:00:00.11
```

```
C:\Users\Dell\Pictures\Saved Pictures\video>
```

Name	Date	Type	Size	Length
 input.mp3	4/9/2026 8:47 PM	MP3 File	51 KB	00:00:12
 output.mp3	4/9/2026 9:28 PM	MP3 File	175 KB	00:00:11

Delay

ffmpeg -i input.mp3 -af "adelay=1000|1000" output.mp3

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp3 -af "adelay=
1000|1000" output.mp3
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
built with gcc 15.2.0 (crosstool-NG 1.28.0.23_185f348)
configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-openccl --enable-gmp --e
nable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
ibb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkvaza
r --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enab
le-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enabl
e-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband --
enable-schannel --enable-sdl2 --enable-libsndio --enable-libsoxr --enable-
libsrt --enable-libsvtav1 --enable-libtwinlame --enable-libuavs3d --disable-l
ibdrm --enable-vaapi --enable-libvidstab --enable-libvenc --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs= --extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags= --cc=x8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm -
-extra-version=20260407
libavutil      60. 29.100 / 60. 29.100
```

```
Stream #0:0: Audio: mp3, 48000 Hz, stereo, fltp
Metadata:
encoder       : Lavc62.29.101 libmp3lame
[out#0/mp3 @ 000001dfde8c9bc0] video:0KiB audio:215KiB subtitle:0KiB other s
treams:0KiB global headers:0KiB muxing overhead: 0.153346%
size=        216KiB time=00:00:13.73 bitrate= 128.6kbits/s speed= 110x elapsed=
0:00:00.12
```

C:\Users\Dell\Pictures\Saved Pictures\video>

Name	Date	Type	Size	Length
 input.mp3	4/9/2026 8:47 PM	MP3 File	51 KB	00:00:12
 output.mp3	4/9/2026 9:29 PM	MP3 File	216 KB	00:00:14

Noise reduction

ffmpeg -i input.mp3 -af "afstdn" output.mp3

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp3 -af "afstdn"
output.mp3
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
  built with gcc 15.2.0 (crosstool-NG 1.28.0.23_185f348)
  configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-openc1 --enable-gmp --e
nable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
ibb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkvaza
ar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enab
le-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enabl
e-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband -
--enable-schannel --enable-sdl2 --enable-libsndio --enable-libsoxr --enable-
libsrt --enable-libsvtav1 --enable-libtwolame --enable-libuavs3d --disable-l
ibdrm --enable-vaapi --enable-libvidstab --enable-libvenc --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs= --extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags= --cc=x8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm -
--extra-version=20260407
  libavutil      60. 29.100 / 60. 29.100
```

```
encoder          : Lavc62.29.101 libmp3lame
[out#0/mp3 @ 000001d4ac61e440] video:0KiB audio:200KiB subtitle:0KiB other s
treams:0KiB global headers:0KiB muxing overhead: 0.165453%
size=          200KiB time=00:00:12.73 bitrate= 128.5kbits/s speed= 101x elapsed=
0:00:00.12
```

C:\Users\Dell\Pictures\Saved Pictures\video>

Name	Date	Type	Size	Length
 input.mp3	4/9/2026 8:47 PM	MP3 File	51 KB	00:00:12
 output.mp3	4/9/2026 9:31 PM	MP3 File	200 KB	00:00:13

Image Compression using cwebp

1. Basic Overview of cwebp

cwebp converts images to **WebP format**, which:

- Reduces size significantly
- Maintains quality
- Improves web performance

Tasks

a) Image Conversion

cwebp input.jpg -o output1.webp

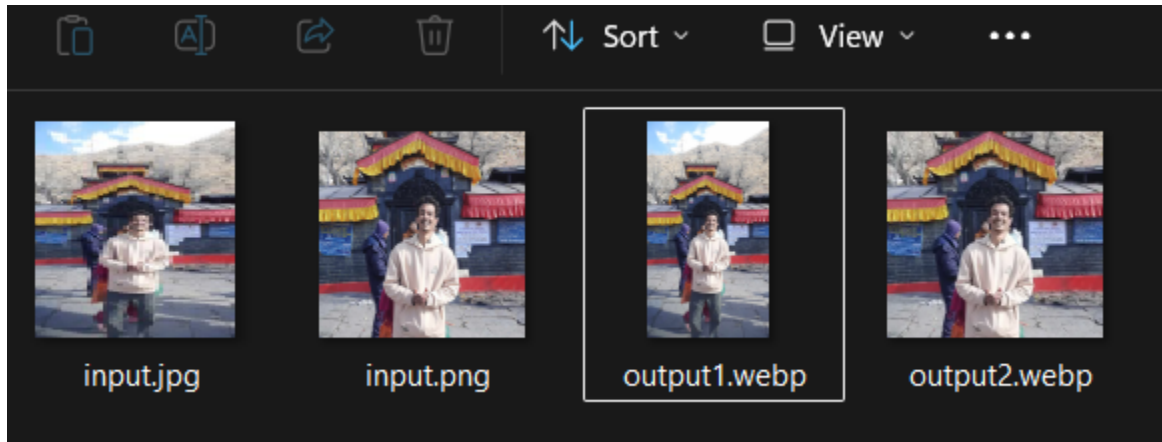
cwebp input.png -o output2.webp

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>cwebp input.jpg -o output1.webp
Saving file 'output1.webp'
File:      input.jpg
Dimension: 1125 x 2000
Output:    365532 bytes Y-U-V-All-PSNR 38.35 46.50 46.73   39.80 dB
           (1.30 bpp)
block count: intra4:      6725  (75.77%)
              intra16:    2150  (24.23%)
              skipped:     495  (5.58%)
bytes used:  header:        455  (0.1%)
              mode-partition: 32436 (8.9%)
Residuals bytes | segment 1 | segment 2 | segment 3 | segment 4 | total
macroblocks:   |      13% |      15% |      40% |      31% |   8875
quantizer:     |      36 |      31 |      25 |      18 |
filter level:  |      11 |       6 |      11 |      21 |

C:\Users\Dell\Pictures\Saved Pictures\video>cwebp input.png -o output2.webp
Saving file 'output2.webp'
File:      input.png
Dimension: 340 x 323
Output:    24980 bytes Y-U-V-All-PSNR 36.70 41.15 41.09   37.74 dB
           (1.82 bpp)
block count: intra4:      458  (99.13%)
              intra16:     4   (0.87%)
              skipped:     0   (0.00%)
bytes used:  header:       163  (0.7%)
              mode-partition: 2457 (9.8%)
Residuals bytes | segment 1 | segment 2 | segment 3 | segment 4 | total
macroblocks:   |      18% |      26% |      34% |      23% |   462
quantizer:     |      36 |      29 |      23 |      16 |
filter level:  |      11 |       6 |       4 |       2 |
```

Name	Date	Type	Size	Tags
 input.jpg	12/21/2023 8:46 PM	JPG File	293 KB	
 input.png	4/9/2026 9:40 PM	PNG File	250 KB	
 output1.webp	4/9/2026 9:42 PM	WEBP File	357 KB	
 output2.webp	4/9/2026 9:42 PM	WEBP File	25 KB	



b) Quality Levels (0–100)

```
cwebp -q 50 input.jpg -o output1.webp
```

```
cwebp -q 90 input.jpg -o output2.webp
```

Output

```

C:\Users\Dell\Pictures\Saved Pictures\video>cwebp -q 50 input.jpg -o output1
.webp
Saving file 'output1.webp'
File:      input.jpg
Dimension: 1125 x 2000
Output:    288946 bytes Y-U-V-All-PSNR 35.60 44.03 44.33   37.07 dB
          (1.03 bpp)
block count:  intra4:      6562  (73.94%)
               intra16:   2313  (26.06%)
               skipped:    465  (5.24%)
bytes used:  header:        363  (0.1%)
               mode-partition: 30796 (10.7%)
Residuals bytes | segment 1 | segment 2 | segment 3 | segment 4 | total
macroblocks:   |      13% |      15% |      40% |      31% |   8875
quantizer:     |      52 |      46 |      37 |      28 |
filter level:  |      16 |      10 |      15 |      49 |

C:\Users\Dell\Pictures\Saved Pictures\video>cwebp -q 90 input.jpg -o output2
.webp
Saving file 'output2.webp'
File:      input.jpg
Dimension: 1125 x 2000
Output:    584322 bytes Y-U-V-All-PSNR 45.12 52.16 52.38   46.48 dB
          (2.08 bpp)
block count:  intra4:      7310  (82.37%)
               intra16:   1565  (17.63%)
               skipped:    480  (5.41%)
bytes used:  header:        771  (0.1%)
               mode-partition: 37957 (6.5%)
Residuals bytes | segment 1 | segment 2 | segment 3 | segment 4 | total
macroblocks:   |      13% |      15% |      40% |      31% |   8875
quantizer:     |      12 |      11 |       8 |       6 |
filter level:  |       4 |       2 |       2 |       0 |

```

Name	Date	Type	Size
 input.jpg	12/21/2023 8:46 PM	JPG File	293 KB
 output1.webp	4/9/2026 9:45 PM	WEBP File	283 KB
 output2.webp	4/9/2026 9:45 PM	WEBP File	571 KB



input.jpg



output1.webp



output2.webp

c) Lossy vs Lossless

Lossy

```
cwebp -q 75 input.png -o output1.webp
```

Lossless

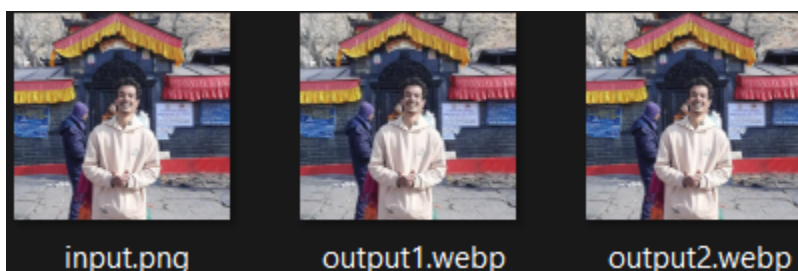
```
cwebp -lossless input.png -o output2.webp
```

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>cwebp -q 75 input.png -o output1
.webp
Saving file 'output1.webp'
File:      input.png
Dimension: 340 x 323
Output:    24980 bytes Y-U-V-All-PSNR 36.70 41.15 41.09   37.74 dB
          (1.82 bpp)
block count:  intra4:      458  (99.13%)
               intra16:    4   (0.87%)
               skipped:    0   (0.00%)
bytes used:  header:      163  (0.7%)
               mode-partition: 2457 (9.8%)
Residuals bytes |segment 1|segment 2|segment 3|segment 4|  total
macroblocks:   |      18%|      26%|      34%|      23%|    462
quantizer:     |      36 |      29 |      23 |      16 |
filter level:  |      11 |       6 |       4 |       2 |

C:\Users\Dell\Pictures\Saved Pictures\video>cwebp -lossless input.png -o out
put2.webp
Saving file 'output2.webp'
File:      input.png
Dimension: 340 x 323
Output:    151506 bytes (11.04 bpp)
Lossless-ARGB compressed size: 151506 bytes
* Header size: 3917 bytes, image data size: 147563
* Lossless features used: PREDICTION CROSS-COLOR-TRANSFORM SUBTRACT-GREEN
* Precision Bits: histogram=3 prediction=3 cross-color=3 cache=0
```

Name	Date	Type	Size	Tags
 input.png	4/9/2026 9:40 PM	PNG File	250 KB	
 output1.webp	4/9/2026 9:48 PM	WEBP File	25 KB	
 output2.webp	4/9/2026 9:49 PM	WEBP File	148 KB	



d) Resize Image

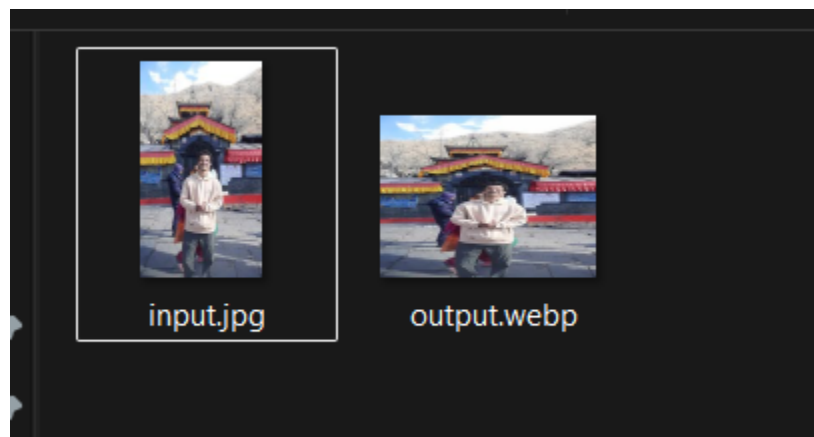
```
cwebp -resize 800 600 input.jpg -o output.webp
```

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>cwebp -resize 800 600 input.jpg
-o output.webp
Saving file 'output.webp'
File:      input.jpg
Dimension: 800 x 600
Output:    81634 bytes Y-U-V-All-PSNR 37.04 43.24 43.28   38.31 dB
          (1.36 bpp)
block count: intra4:      1697  (89.32%)
              intra16:    203  (10.68%)
              skipped:     62  (3.26%)
bytes used: header:        219  (0.3%)
              mode-partition: 8691 (10.6%)
Residuals bytes | segment 1 | segment 2 | segment 3 | segment 4 | total
macroblocks:   |    19%   |    18%   |    39%   |    23%   |  1900
quantizer:     |    36   |    30   |    23   |    17   |
filter level:  |    11   |     6   |     4   |    16   |

C:\Users\Dell\Pictures\Saved Pictures\video>
```

Name	Date	Type	Size
 input.jpg	12/21/2023 8:46 PM	JPG File	293 KB
 output.webp	4/9/2026 9:51 PM	WEBP File	80 KB



e) JPEG vs WebP Comparison

Feature	JPEG	WebP
File Size	Larger	Smaller
Quality	Good	Better at same size
Transparency	Not supported	Supported
Web Speed	Slower	Faster

Conclusion

FFmpeg is a powerful tool for multimedia processing that supports a wide range of operations including conversion, editing, filtering, and compression. The use of cwebp improves image optimization significantly, making both tools essential for modern multimedia and web development.

Lab 8: Write a program to implement a data compression algorithm.

Objective: To understand and implement basic data compression algorithms such as Run-Length Encoding (RLE) and Huffman Encoding.

Introduction: Data compression is a technique used to reduce the size of data for efficient storage and transmission. It can be lossless (no data loss) or lossy.

Algorithms Implemented:

Run-Length Encoding (RLE)

Concept: RLE compresses data by replacing repeated characters with a single character and its count.

Example:

Input: AAAABBBCCDAA

Output: A4B3C2D1A2

Python Program:

```
def rle_encode(data):
    encoding = ""
    i = 0

    while i < len(data):
        count = 1
        while i + 1 < len(data) and data[i] == data[i + 1]:
            i += 1
            count += 1
        encoding += data[i] + str(count)
        i += 1

    return encoding
data = "AAAABBBCCDAA"
print("Encoded:", rle_encode(data))
```

Huffman Encoding

Concept: Huffman Encoding is a lossless compression technique that assigns shorter codes to frequent characters and longer codes to less frequent ones.

Steps:

1. Calculate frequency of characters
2. Build a binary tree
3. Assign codes
4. Encode data

Python Program:

```
import heapq
from collections import Counter

class Node:
    def __init__(self, char, freq):
        self.char = char
        self.freq = freq
        self.left = None
        self.right = None

    def __lt__(self, other):
        return self.freq < other.freq

def build_huffman_tree(text):
    frequency = Counter(text)
    heap = [Node(char, freq) for char, freq in frequency.items()]
    heapq.heapify(heap)

    while len(heap) > 1:
        left = heapq.heappop(heap)
        right = heapq.heappop(heap)

        merged = Node(None, left.freq + right.freq)
        merged.left = left
        merged.right = right
```

```

    heapq.heappush(heap, merged)

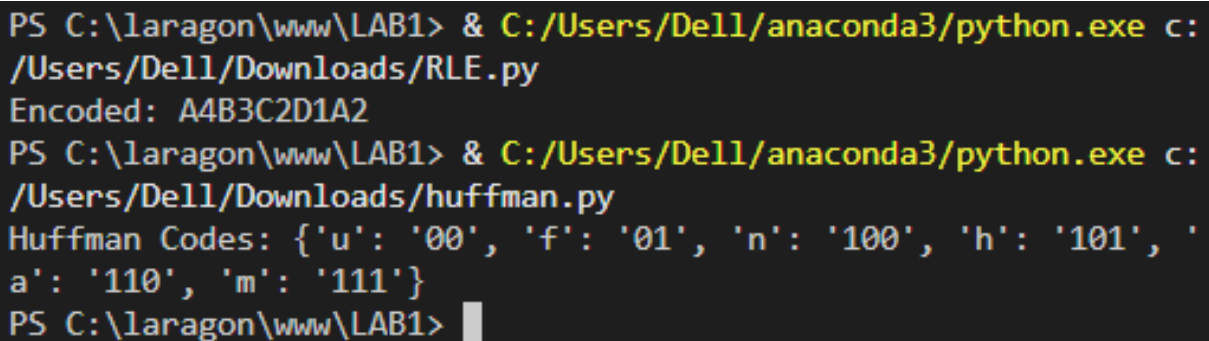
return heap[0]

def generate_codes(node, prefix="", codebook={}):
    if node:
        if node.char:
            codebook[node.char] = prefix
            generate_codes(node.left, prefix + "0", codebook)
            generate_codes(node.right, prefix + "1", codebook)
        return codebook
text = "huffman"
tree = build_huffman_tree(text)
codes = generate_codes(tree)

print("Huffman Codes:", codes)

```

Output:



```

PS C:\laragon\www\LAB1> & C:/Users/Dell/anaconda3/python.exe c:
/Users/Dell/Downloads/RLE.py
Encoded: A4B3C2D1A2
PS C:\laragon\www\LAB1> & C:/Users/Dell/anaconda3/python.exe c:
/Users/Dell/Downloads/huffman.py
Huffman Codes: {'u': '00', 'f': '01', 'n': '100', 'h': '101', '
a': '110', 'm': '111'}
PS C:\laragon\www\LAB1>

```

Conclusion:

Run-Length Encoding is simple and effective for repetitive data, while Huffman Encoding provides efficient compression by using variable-length codes. Both techniques are important in data compression and real-world applications.